

Light Energy

Use the cue, then fade the cue. Study the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

CUE FOCUS Light travels in **straight lines**. It can **reflect** (bounce off a mirror), **refract** (bend when it passes into water, making a straw look bent), or be **absorbed or blocked** (making a shadow).

Round 1 — Retrieve It

1. How does light travel — in straight lines or in curves?

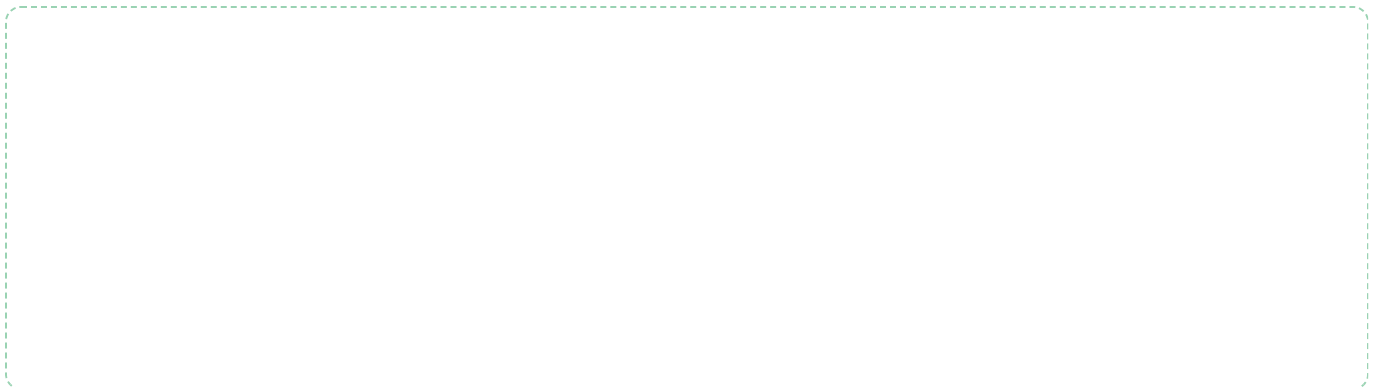
2. What do we call it when light bounces off a mirror?

3. What do we call it when light bends as it passes into water?

4. What forms when an object blocks light?

Round 2 — Sketch It

Draw a straw sitting in a clear glass of water. Show how the straw looks bent where it enters the water. Label the part where the light refracts (bends).



Light Energy — Apply It

Keep the cue card covered. Answer first. Then use the cue to check, revise, and explain what you changed.

Round 3 — Apply the Idea

1. You shine a flashlight at a mirror and the light lands on the ceiling. Which behavior of light made that happen?

2. A pencil in a glass of water looks broken at the water line, even though it is not. What behavior of light explains this?

3. You stand outside on a sunny day and see a dark shape of yourself on the ground. Why does your body make that shape?

Round 4 — Reason from Evidence

A student holds a solid book in front of a lamp and a clear window in front of the same lamp. The book makes a dark shadow, but the window barely makes a shadow at all. Use what you know about light to explain the difference.

Light Energy — ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A

Articulate

Explain the difference between light reflecting and light refracting. Give one example of each.

C

Connect

Connect one way light behaves to something you have seen in real life. Name the behavior and describe what you saw.

E

Extend

Plan a way to make a shadow larger and then smaller using one flashlight and one object. Explain what you would move and why.

Confidence check. Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it.

Teacher Key — Light Energy

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. In straight lines.
2. Reflection (the light reflects).
3. Refraction (the light refracts, or bends).
4. A shadow.

Sketch It — look for

Straw drawn looking bent or shifted at the water line, with the bend labeled where the light refracts as it passes into the water.

Apply It

1. Reflection — the light bounced off the mirror.
2. Refraction — light bends as it passes from air into water, so the pencil looks broken.
3. Your body blocks (absorbs and stops) the light, so no light reaches the ground behind you, making a shadow.

Reason from Evidence — look for

The book is not clear, so it blocks the light and makes a dark shadow. The window is clear, so light passes through it and almost no light is blocked, making little or no shadow.

ACE — Extend look-for

Plan changes the distance between the object and the flashlight (or the object and the wall). Moving the object closer to the light makes a bigger shadow; moving it farther from the light makes a smaller shadow.

Suggested use. Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color.